Question-5.2.25

EE25BTECH11022 - sankeerthan

Question

Solve the following system of linear equations $ax + by = c$, $bx + ay = 1 + c$

Solution

Given

$$ax + by = c \tag{1}$$
$$bx + ay = 1 + c \tag{2}$$

The matrix equation for a line is defined as

$$\mathbf{n}^\top \mathbf{x} = c \tag{3}$$

where $\mathbf{n}$ is the coefficient matrix and $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{pmatrix} a & b \end{pmatrix} \mathbf{x} = c \tag{4}$$
$$\begin{pmatrix} b & a \end{pmatrix} \mathbf{x} = 1 + c \tag{5}$$

Solution

As a matrix equation

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} c \\ 1+c \end{pmatrix} \tag{6}$$

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}^\top \begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}^\top \begin{pmatrix} c \\ 1+c \end{pmatrix} \tag{7}$$

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} c \\ 1+c \end{pmatrix} \tag{8}$$

$$\begin{pmatrix} a^2+b^2 & 2ab \\ 2ab & a^2+b^2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} b+(a+b)\,c \\ a+(a+b)\,c \end{pmatrix} \tag{9}$$

solution

$$\mathbf{x} = \frac{1}{(a^2-b^2)^2}\begin{pmatrix} a^2+b^2 & -2ab \\ -2ab & a^2+b^2 \end{pmatrix}\begin{pmatrix} b+(a+b)\,c \\ a+(a+b)\,c \end{pmatrix} \tag{10}$$

$$\mathbf{x} = \frac{1}{(a^2-b^2)^2}\begin{pmatrix} \left(a^2+b^2\right)\left(b+(a+b)\,c\right)-2ab\left(a+(a+b)\,c\right) \\ -2ab\left(b+(a+b)\,c\right)+\left(a^2+b^2\right)\left(a+(a+b)\,c\right) \end{pmatrix} \tag{11}$$

$$\mathbf{x} = \frac{1}{(a^2-b^2)^2}\begin{pmatrix} b\left(a^2+b^2-2a^2\right)+\left((a+b)\,c\right)\left(a^2+b^2-2ab\right) \\ \left((a+b)\,c\right)\left(a^2+b^2-2ab\right)+a\left(a^2+b^2-2b^2\right) \end{pmatrix} \tag{12}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{(a^2-b^2)^2}\begin{pmatrix} \left((a+b)\,c\right)(a-b)^2-b\left(a^2-b^2\right) \\ \left((a+b)\,c\right)(a-b)^2+a\left(a^2-b^2\right) \end{pmatrix} \tag{13}$$

$$\tag{14}$$

Hence $x = \frac{c}{a+b} + \frac{b}{b^2-a^2}, y = \frac{c}{a+b} + \frac{a}{a^2-b^2}$ is the solution for given system of linear equations

graph

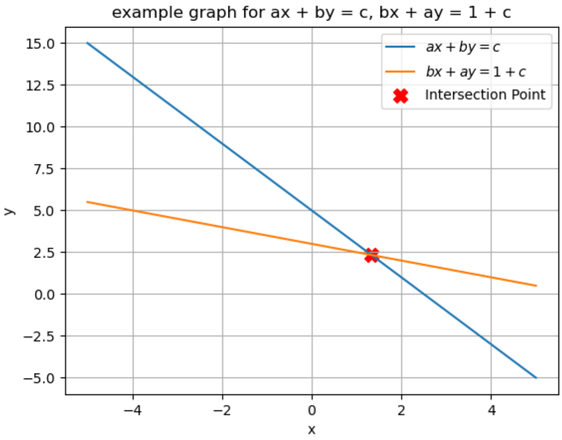

Figure:

C-Code

```c
// code.c
#include <stdio.h>

// Compute (x, y) for system:
// ax + by = c
// bx + ay = 1 + c

void solve_system(double a, double b, double c, double *x, double
     *y) {
    double denom = a*a - b*b;
    *x = (a*c - b*(1.0 + c)) / denom;
    *y = (-b*c + a*(1.0 + c)) / denom;
}
```

C-Code

```c
// Fills arrays with points on the line for fixed a, b, c, and
    range of x values
void generate_points(double a, double b, double c, double *xs,
    double *ys, int n) {
    double x_min = -5.0, x_max = 5.0;
    for(int i=0; i<n; ++i) {
        xs[i] = x_min + (x_max - x_min)*i/(n-1);
        ys[i] = (c - a*xs[i]) / b;
    }
}
```

Python-Code

```python
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

lib = ctypes.CDLL('./code.so')
```

Python-Code

```python
# Function signatures
lib.solve_system.argtypes = [ctypes.c_double, ctypes.c_double,
    ctypes.c_double,
                              ctypes.POINTER(ctypes.c_double), ctypes
                                  .POINTER(ctypes.c_double)]
lib.generate_points.argtypes = [ctypes.c_double, ctypes.c_double,
     ctypes.c_double,
                                np.ctypeslib.ndpointer(dtype=np.
                                    float64, flags="C_CONTIGUOUS"),
                                np.ctypeslib.ndpointer(dtype=np.
                                    float64, flags="C_CONTIGUOUS"),
                                ctypes.c_int]

def solve_system(a, b, c):
    x = ctypes.c_double()
    y = ctypes.c_double()
    lib.solve_system(a, b, c, ctypes.byref(x), ctypes.byref(y))
    return x.value, y.value
```

Python-Code

```python
def extract_points(a, b, c, n=20):
    xs = np.zeros(n, dtype=np.float64)
    ys = np.zeros(n, dtype=np.float64)
    lib.generate_points(a, b, c, xs, ys, n)
    return xs, ys

if __name__ == "__main__":
    a, b, c = 2, 1, 5
    xsol, ysol = solve_system(a, b, c)
    print("x =", xsol, "; y =", ysol)
    xs, ys = extract_points(a, b, c, 10)
    print("xs:", xs)
    print("ys:", ys)
```

Python-Code

```
# plot.py
import numpy as np
import matplotlib.pyplot as plt
from call import solve_system

# Parameters
a, b, c = 2, 1, 5

# Solve for intersection
x_int, y_int = solve_system(a, b, c)

# Generate x values for plotting
x_vals = np.linspace(-5, 5, 200)

# Calculate y-values for each line
# Line 1: ax + by = c => y = (c - a*x) / b
y_line1 = (c - a * x_vals) / b
```

Python-Code

```
# Line 2: bx + ay = 1 + c => y = (1 + c - b*x) / a
y_line2 = (1 + c - b * x_vals) / a

# Plot lines
plt.plot(x_vals, y_line1, label=r'$ax + by = c$')
plt.plot(x_vals, y_line2, label=r'$bx + ay = 1 + c$')

# Plot intersection point
plt.scatter([x_int], [y_int], color='red', marker='X', s=100,
    label='Intersection Point')

plt.xlabel('x')
plt.ylabel('y')
plt.title('example graph for ax + by = c, bx + ay = 1 + c')
plt.grid(True)
plt.legend()
plt.show()
```